

Meeting 1 - 03/07/2026

LEO Satellite Emulator — Meeting Notes

Project Overview

The project aims to develop an **emulation tool for LEO satellite environments**.

The proposed tool will integrate:

- Satellites represented as **containers**
- **Energy modelling**, including batteries and solar panels
- **Mininet-based networking**
- **Fixed and mobile ground stations**
- **Satellite trajectory/orbit modelling**

The long-term goal is to develop an **open-source tool** that can support further research.

Research Motivation

Existing tools should be reviewed to determine whether there is a gap in combining:

- Container-based satellite environments
 - Mininet networking
 - Energy/solar modelling
 - Mobile ground stations
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Immediate Research Focus

Review of existing **LEO satellite simulators and emulators**.

The review should identify:

- Existing tools

- Their main capabilities
- Their underlying approaches
- Networking support
- Energy/solar modelling
- Ground-station modelling
- Mobility and trajectory modelling
- Container support
- Limitations and research gaps

The results will be organised into a comparison table.

Background Research

Relevant background areas include:

- LEO orbits and satellite movement
 - Satellite networking
 - Ground stations
 - Satellite energy generation and storage
 - Solar panels and batteries
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Related Work

Georgia Tech

closely related **Georgia Tech LEO satellite research** to understand the scope and existing approaches in this area.

CloudSim

CloudSim as an important reference for simulation and modelling in this area.

Long-Term Direction

Following the literature review:

1. Identify limitations and research gaps in existing tools.

2. Define the proposed tool's contribution.
 3. Develop the emulator.
 4. Validate the emulator through a simple experiment.
 5. Use the tool for further research, such as:
 - Battery optimisation
 - Task offloading between satellites
 - Network optimisation
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Potential Tech Stack

- **Docker**
- **Mininet**
- **Bash/Linux**
- **Mathematical trajectory modelling**
- **Energy/solar modelling**